

AgriSmart AI: Team Handoff & Next Steps Update

Hey team! 🙌 The repository has been cleaned up, updated, and pushed to GitHub. Since you are all added as collaborators, here is exactly how to get started!

HOW TO SETUP LOCALLY (For Beginners)

1. **Clone the Repo:** `git clone https://github.com/Vatsal728/AgriMartAI.git`
2. **Create your working branch:** Please don't push straight to main! Make your own branch: `git checkout -b <your-name>-feature`
3. **Run the Setup:** We have automated this! Just double-click the `setup_env.bat` file in the root folder. It will install all requirements.
4. **Run the App:** Double-click the `run_dashboard.bat` file to start the Streamlit AI interface.

✅ WHAT'S COMPLETED & PUSHED TO `main`:

1. **Clean Repository:** Removed all obsolete model files, draft PDFs, and temp images.
 2. **AI Vision Models (Included in Git):**
 - `models/efficientnet_b0_best.pth` (46.8 MB) → Primary model for **Web API / Cloud Backend** (99.8% Acc).
 - `models/mobilenet_v3_best.pth` (18.0 MB) → Lightweight model for **Mobile App** (99.5% Acc).

(Both models are < 100MB and tracked in Git — no extra downloading needed!)
 3. **Unified Chatbot Prototype:** Works seamlessly with single image upload + custom question text simultaneously.
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NEXT STEPS (MIGRATION TO PRODUCTION 10/10):

Backend Team (FastAPI):

- Migrate `src/cv_pipeline/predict.py` and `src/advisor/agentive_advisor.py` into FastAPI REST endpoints (`/api/v1/diagnose`, `/api/v1/chat`, `/api/v1/telemetry`).
- Connect persistent database (PostgreSQL/SQLite) for user farm profiles and persistent chat memory.

Frontend Team (React / Next.js):

- Build a sleek, modern ChatGPT-style web UI using **React (Next.js) + TailwindCSS + Shadcn UI**.
- **Important:** Make sure to look at the reference Git repo we shared earlier for the exact design/layout you should mirror!
- Create dashboard cards for live satellite weather (Temperature, Humidity, Rain Risk, Soil Moisture).



Data & RAG Team (Claude Pro Task):

- **Whoever has the Claude Pro subscription**, your task is to generate and compile the extra data we need for a 10/10 RAG!
- Map our generic chemical names (e.g. *Mancozeb 75% WP*) to local commercial brand names and estimated costs per acre.
- Add Hindi/Regional language translations for Q&A responses.

👉 **Action Item:** Follow the setup steps above, pull the code, and let's start the FastAPI + React sprint! 🔥